

In this paper, all exercises in Hoffman / Kunze will be solved in complete form.

Sec. 3.5 linear functional.

#3.5.3. For any $A, B \in M_{n \times n}(F)$, $\text{trace}(AB) = \text{trace}(BA)$.

Moreover, if $A \sim B$, then $\text{trace } A = \text{trace } B$.

Proof. Since $\text{trace } AB = \sum_{j=1}^n \sum_{k=1}^n A_{jk} B_{kj}$ and $\text{trace } BA = \sum_{j=1}^n \sum_{k=1}^n B_{jk} A_{kj}$

so just rename the indices, we obtain the first assertion.

Now, suppose that $A = U^{-1}BU$ for some $U \in GL(F)$. Now,

$$\text{trace } A = \text{trace } U^{-1}BU = \text{trace } BUU^{-1} = \text{trace } B.$$

Sec 6.2.

6.2.1. Note f_2 is the characteristic polynomial of the given matrix A_2 .

$$f_1(t) = t(t-1), \quad f_2(t) = (2-t)/(1-t) + 3 = t^2 - 3t + 5, \quad f_3(t) = (1-t)^2 - 1 = t^2 - 2t$$

In $\mathbb{R}$, f_2 and f_3 do not split over the given field, actually there are no solution in $\mathbb{R}$ of f_2 and f_3 . So A_2 and A_3 have no eigenvalue.

But, $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $E_0 = \text{Ker } A = \text{span}_{\mathbb{R}} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

And $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, so $E_1 = \text{span}_{\mathbb{R}} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$, $\frac{2 \pm \sqrt{4-8}}{2}$

In $\mathbb{C}^2$, A_1 has same eigenvectors. But:

$$t = \frac{3 \pm \sqrt{9-20}}{2} = \frac{3 \pm i\sqrt{11}}{2} \text{ are roots of } f_2, \text{ and}$$

$$t = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i \text{ are roots of } f_3. \text{ so,}$$

$$A_2 - \underbrace{\left(\frac{3 \pm i\sqrt{11}}{2} \right)}_{\lambda_2} I = \begin{bmatrix} \frac{1}{2} \pm i \frac{\sqrt{11}}{2} & 3 \\ -1 & -\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \end{bmatrix}$$

$$\text{Then, } (A_2 - \lambda_2)I \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \left(\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right)x + 3y \\ -x + \left(-\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right)y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} \left(\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right)x + 3y = 0 \\ \left(\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right)x + 3y = 0. \end{cases}$$

(multiplying $\left(\frac{1}{2} - i \frac{\sqrt{11}}{2} \right)$ to the bottom)

so, the eigenvectors are:

$$\begin{pmatrix} -1 \\ \frac{1}{3} \left(\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right) \end{pmatrix}, \text{ corresponding to } \frac{1}{2} \pm i \frac{\sqrt{11}}{2}$$

#6.2.6. $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{bmatrix}$ over $\mathbb{R}$.

$\det(A - tI) = t^4$, and A has the only eigenvalue as 0. So,

A is diagonalizable iff $\dim \ker A = 4$.

But, the only possibility for $\dim \ker A = 4$ is $a=b=c=0$, because if at least one of a or b or c is not zero, then $\text{rank } A \geq 1$.

#6.2.7. If $T: F^n \rightarrow F^n$ has n distinct eigenvalues, then T is diagonalizable.

Proof Let $\alpha_1, \dots, \alpha_n \in F^n$ be the eigenvectors corresponding to distinct eigenvalues. Then, $\{\alpha_1, \dots, \alpha_n\}$ are linearly independent, because: Suppose that

$$c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n = 0 \text{ where } c_i \in F.$$

Then, for any F -polynomial $f(t) \in F[t]$,

$$0 = f(T)(c_1 \alpha_1 + \dots + c_n \alpha_n) = f(c_1) \cdot \alpha_1 + \dots + f(c_n) \alpha_n. \quad (*)$$

$$\text{Take } f_i(t) = \prod_{j \neq i} (t - c_j) / \prod_{j \neq i} (c_i - c_j)$$

Then, $f_i(c_i) = 1$, so, we can conclude $c_1 = c_2 = \dots = c_n = 0$,

by substituting $f_1, \dots, f_n$ to $(*)$.

#6.2.8. Let $A, B \in M_n(F)$. If $I - AB$ is invertible, then $I - BA$ is invertible with

$$(I - BA)^{-1} = I + B \cdot (I - AB)^{-1} \cdot A$$

Proof. Let $C = (I - AB)^{-1}$. Then,

$C - ABC = C - CAB = I$, so $ABC = CAB$ (That is, C and AB commute.)

$$\begin{aligned} \text{Now } (I - BA) \cdot (I + BCA) &= I + BCA - BA - BABC A \\ &= I + BCA - BA - B(C - I)A \\ &= I + BCA - BA - BCA + BA \\ &= I. \end{aligned}$$

So, proved.

#6.2.9. Using the fact above, deduce that

AB and BA have precisely the same eigenvalue in F .

Proof. If $t = 0$, then

t is an eigenvalue for $AB \Leftrightarrow \det AB = 0 \Leftrightarrow \det BA = 0 \Leftrightarrow t \text{ is an eigenvalue for } BA$.

$$\begin{aligned} \text{If } t \neq 0, \text{ then } \det(AB - tI) = 0 &\Leftrightarrow t^n \cdot \det((AB) \cdot t^{-1} - I) = 0 \\ &\Leftrightarrow \det((BA) \cdot t^{-1} - I) = 0 \\ &\Leftrightarrow \det(BA - tI) = 0. \end{aligned}$$

So, proved.

#6.2.10. A 2×2 symmetric matrix is diagonalizable.

Proof. Let $A = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ be given. Then

$$\det(A - tI) = (t-a)(t-b) - c^2 = t^2 - (a+b)t + ab - c^2$$

If $c=0$, then it is clearly diagonal. And, if $c \neq 0$, then $c^2 > 0$, so the characteristic polynomial has two distinct roots r_1 and r_2 ,

$$\dim \ker(A - r_1 I) \geq 1 \quad \text{and} \quad \dim \ker(A - r_2 I) \geq 1,$$

so $\dim \ker(A - r_1 I) = \dim \ker(A - r_2 I) = 1$, so diagonalizable.

#6.2.11. If $N \in M_{2 \times 2}(\mathbb{C})$ satisfies $N^2 = 0$, then either $N=0$ or $N \sim \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

Proof. If $N=0$, then we are done. Suppose that $N \neq 0$, but $N^2=0$.

(Directly, the minimal polynomial of N is t^2 , so it can be triangular.

But in this, we prove more specifically.)

Since $N^2=0$ implies $N \cdot Nx = 0$ for all $x \in \mathbb{C}^2$, so

$$\operatorname{Im} N \subseteq \ker N.$$

Thus $\ker N \neq \{0\}$, take $d_1 \in \ker N$ be non-zero.

And, since $N \neq 0$, $\ker N \neq \mathbb{C}^2$. So take $d_2 \in \mathbb{C}^2 \setminus \ker N$ s.t. $Nd_2 \in \ker N$.

Now, $Nd_1 = 0$ and $Nd_2 = C \cdot d_1$ for some non-zero scalar $C \in \mathbb{C}$, so

$$[L_N]_{\{d_1, d_2\}} = \begin{bmatrix} 0 & C \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad [L_N]_{\{d_2, d_1\}} = \begin{bmatrix} 0 & 0 \\ C & 0 \end{bmatrix}.$$

(To conclude the conversation completely, take $d_2' = C^{-1}d_2$).

#6.2.12. Every Complex 2×2 Matrix is similar to $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ or $\begin{bmatrix} a & 1 \\ 0 & a \end{bmatrix}$.

Proof. The characteristic polynomial of the given Matrix must be degree 2 complex polynomial.

So, there are only two possibilities: the polynomial has only one root of multiplicity 2

or distinct two roots.

In the second case, it is diagonalizable. In the first case, let $C \in \mathbb{C}$ be the root.

Then, $(N - CI)^2 = 0$, because $\det(N - tI) = (t-C)^2$, so $\ker(N - CI) = \mathbb{C}^2$.

So $N - CI \sim \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, or $N \sim \begin{bmatrix} a & 1 \\ 0 & a \end{bmatrix}$.

#6.2.13.

Let $V = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$, and define

$$T: V \rightarrow V; f \mapsto \int_0^x f(t) dt.$$

Then T has no eigenvalues.

Proof. Since the integral has linearity, T is a linear map.

Suppose that T has an eigenvalue, $c \in \mathbb{R}$. That is,

$$T(f) = c \cdot f(x) = \int_0^x f(t) dt \quad \text{for some non-zero } f \in V.$$

If $c=0$, then $\int_0^x f(t) dt = 0$ for all x , so f is zero, contradiction.

If $c \neq 0$, then $c \cdot f(x) = \int_0^x f = 0$. Moreover,

$$c \cdot F'(x) = F(x) \Rightarrow F'(x) = c^{-1} e^{cx} \Rightarrow f(x) = e^{cx}$$

But $e^{c0} = e^0 = 1 \neq 0$, so contradiction.

#6.2.14.

Let $A \in M_{n \times n}(F)$ be a matrix with ch. polynomial

$$(x - c_1)^{d_1} \dots (x - c_k)^{d_k}$$

Let $V = \text{span}_F \{B \in M_{n \times n} \mid AB = BA\}$, then $\dim V = d_1^2 + \dots + d_k^2$.

Proof.

#6.3.2.

Let $a, b, c \in F$. Then,

$$A = \begin{bmatrix} 0 & 0 & -c \\ 1 & 0 & -b \\ 0 & 1 & -a \end{bmatrix} \text{ has characteristic polynomial as } x^3 + ax^2 + bx + c, \text{ and it is minimal polynomial for } A.$$

Proof. $\det(xI - A) = \begin{vmatrix} x & 0 & c \\ -1 & x & b \\ 0 & -1 & x+a \end{vmatrix} = x \cdot (x(x+a) + b) + c = x^3 + ax^2 + bx + c.$

And, $A^2 = \begin{bmatrix} 0 & -c & ac \\ 0 & -b & -c+ab \\ 1 & -a & -b+a^2 \end{bmatrix}$ and $A^3 = \begin{bmatrix} -c & ac & bc-a^2c \\ -b & -c+ab & ac+b^2-a^2b \\ -a & -b+a^2 & -c+2ab-a^3 \end{bmatrix}$

So, by the just calculation, $A^3 + aA^2 + bA + cI = 0$.

Now, $\{I, A, A^2\}$ is linearly independent, so there is no polynomial p of degree 2 s.t.

$$p(A) = 0$$

So, the polynomial $x^3 + ax^2 + bx + c$ is minimal polynomial for A .

#6.3.3. Let $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ -2 & -2 & 2 & 1 \\ 1 & 1 & -1 & 0 \end{bmatrix}$. Then, $\det(xI - A) =$

#6.3.5. Let V be an n -dimensional Vector space and $T: V \rightarrow V$ be a linear operator. If $T^k = 0$ for some $k \in \mathbb{N}$, then $T^n = 0$.

Proof. If $k \leq n$, then clearly, $T^n = T^{n-k} \circ T^k = 0$.

Suppose that $k > n$. By the Cayley-Hamilton Theorem, the minimal polynomial $p(x)$ of T satisfies $\deg p(x) \leq n$, since $p(x)$ divides the characteristic polynomial of T .

So, the minimal polynomial of T also divides T^k , so $T^r = 0$ for some $r \leq n$.

Now, $T^n = T^{n-r} \circ T^r = 0$.

#6.3.6. Find $\begin{smallmatrix} a \\ 3 \end{smallmatrix}$ by 3 Matrix for which the minimal polynomial is x^2 .

Proof. A has minimal polynomial as x^2 if and only if:

$A^2 = 0$ and $\{A, I\}$ is linearly independent.

So, $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

#6.3.7. Let $D: P_n(\mathbb{R}) \rightarrow P_n(\mathbb{R})$ be the differentiation operator.

Evaluate the minimal polynomial for D .

Proof. For any polynomial $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \in P_n(\mathbb{R})$,

$$D^n(\sum a_i x^i) = n! \cdot a_n$$

And, so $D^{n+1} = 0$, the zero map. Now, suppose that

$$C_n D^n + C_{n-1} D^{n-1} + \dots + C_1 D + C_0 I = 0. \text{ where } C_0, \dots, C_n \in \mathbb{R}.$$

Subtracting Polynomials:

$$f_0(x) = 1, f_1(x) = x, f_2(x) = \frac{1}{2!} x^2, f_3(x) = \frac{1}{3!} x^3, \dots, f_n(x) = \frac{1}{n!} x^n$$

Then, the all coefficients are zero, so $\{1, D, \dots, D^n\}$ is linearly independent.

In other say, the polynomial of degree less than $n+1$ cannot be the minimal polynomial.

+Comment for 6.3.7. $\mathbb{R}$ 이 아니라 임의의 체 F 에서 가능, 하지만 $\text{char}(F) = 0$ 이어야 함
 $\mathbb{Z}$ 이 특이한 예외일 것 같으나, 3/13/2020.

#6.3.8.

Since $p^2 = I$, the minimal polynomial is $x^2 - x = x(x-1)$.

#6.3.9. If the characteristic polynomial of $A \in M_{n \times n}(F)$ is

$$f = (x - c_1)^{d_1} \cdots (x - c_k)^{d_k},$$

then $\text{trace } A = c_1 d_1 + \cdots + c_k d_k$.

Proof. Since $f = x^n - (c_1 d_1 + \cdots + c_k d_k) \cdot x^{n-1} + \cdots$,

therefore $\text{trace } A = c_1 d_1 + \cdots + c_k d_k$.

Sec 6.4.

#6.4.1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix} \in M_{2 \times 2}(\mathbb{R})$.

a). The L_A -invariant subspaces are only $\mathbb{R}^2$ and $\{0\}$.

Proof. Let $\{0\} < W < \mathbb{R}^2$ be a L_A -invariant subspace.

Then, $W = \text{span}_{\mathbb{R}}(a, b)$ for some non-zero vector $(a, b) \in \mathbb{R}^2$. So, by the assumption,

$$A \cdot \begin{pmatrix} a \\ b \end{pmatrix} \in W, \text{ or } A \begin{pmatrix} a \\ b \end{pmatrix} = c \cdot \begin{pmatrix} a \\ b \end{pmatrix} \text{ for some } c \in \mathbb{R}.$$

But, $\det(xI - A) = (x-1)(x-2) + 2 = x^2 - 3x + 4 = (x - \frac{3}{2})^2 + \frac{7}{4}$ has no solutions in $\mathbb{R}$, so contradiction.

But, If A is a matrix in $\mathbb{C}$, then it has an eigenvalue α , so $\text{span}_{\mathbb{C}}\{\alpha\}$ is the desired.

#6.4.2. Let W be an invariant space for T .

Then, the minimal polynomial of $T|_W$ divides the minimal polynomial of T .

Proof. Let $p(x)$ be the minimal polynomial of T . Then, for any $\alpha \in W$,

$$p(T)(\alpha) = p(T|_W)(\alpha) = 0.$$

So, $p(T|_W) = 0$, thus we obtain the result.

#6.4.3. Let $c \in F$ be an eigenvalue of T .

For all $\alpha \in \ker(T - cI)$, $T(\alpha) = c \cdot \alpha$, so

$$T|_{\ker(T-cI)} = c \cdot I; \ker(T-cI) \rightarrow \ker(T-cI); \alpha \mapsto c\alpha.$$

So, the restriction is c -multiple of Identity map if $c \neq 0$,
or zero map if $c = 0$.

#6. 4, 4.

Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 2 & -2 & 2 \\ 2 & -3 & 2 \end{bmatrix}$ over $\mathbb{R}$.

$$\det(xI - A) = \begin{vmatrix} x & -1 & 0 \\ -2 & x+2 & -2 \\ -2 & 3 & x-2 \end{vmatrix} = x((x+2)(x-2) + 6) + (-2)(x-2) - 4$$
$$= x(x^2 + 2) - 2x$$
$$= x^3$$

Let $\alpha = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$. Then, $A\alpha = 0$.